Name

Mathematics Stage 5 Core

Expanding Expressions

Book 1

Version: 261006

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-ALG-C-01 simplifies algebraic fractions with numerical denominators and expands algebraic expressions

Apply the distributive law to the expansion of algebraic expressions, and collect like terms where appropriate

- Expand algebraic expressions, including those with negative coefficients
- Expand and simplify algebraic expressions by removing grouping symbols and collecting like terms where appropriate
- Expand binomial products algebraically using the distributive law or consider areas of rectangles as a possible method of expansion

Expanding Brackets with Negative Coefficients

Review

1 Expand brackets

a	$7(l+4)$	**b**	$4(2x-6)$	**c**	$5(3e+8)$
d	$9(x-9)$	**e**	$10(6g-7)$	**f**	$7(8x-2)$
g	$6i(t-v)$	**h**	$7y(4y+7g)$	**i**	$14k(k-3)$

Expanding Brackets

To expand brackets, we use the distributive law:

$$a(b+c) = ab + ac$$

Example Expand brackets using the distributive law

Expand:

$-x(2y - z)$ $= -2xy + xz$	To expand a bracket, multiply e…… term inside the brackets by the term o……

Guided Practice

Expand:

a $-3(m+3)$ $-3m - 9$	**b** $-(y+4)$ $-y - 4$	**c** $-5(2m+7)$ $-10m - 35$
d $-x(8-3m)$ $-8x + 3xm$	**e** $-y(y-9)$ $-y^2 + 9y$	**f** $-4t(t-8)$ $-4t^2 + 32t$

Key Ideas

1. The distributive law says to expand brackets by …… the term on the outside by the terms on this inside of the brackets.
2. In the expression $-x(a+b)$, we multiply each term inside the brackets by ……

1 Expand these expressions.

a $3(x+4)$

$3x+12$

b $3(4+x)$

$12+3x$

c $3(4-x)$

$12-3x$

d $3(x-4)$

$3x-12$

e $3(-x-4)$

$-3x-12$

f $-3(x+4)$

$-3x-12$

g $-3(x-4)$

$-3x+12$

h $-3(2x-4)$

$-6x+12$

i $-3(4-2x)$

$-12+6x$

j $3(x+4)$

$3x+12$

k $-6(2-x)$

$-12+6x$

l $6(x-2)$

$6x-12$

m $6(x-2y)$

$6x-12y$

n $6(5x-2y)$

$30x-12y$

o $-6(2y-5x)$

$-12y+30x$

p $-6(2y-5x-7z)$

$-12y+30x+42z$

q $-6(5x-2y-7z)$

$-30x+12y+42z$

r $-6(-5x-2y-7z)$

$30x+12y+42z$

s $-(-5x-2y-7z)$

$5x+2y+7z$

t $-6(2-x)$

$-12+6x$

2 Expand these expressions.

a $x(x+4)$

x^2+4x

b $x(4+x)$

$4x+x^2$

c $x(4-x)$

$4x-x^2$

d $x(x-4)$

x^2-4x

e $x(-x-4)$

$-x^2-4x$

f $-x(x+4)$

$-x^2-4x$

g $-x(x-4)$

$-x^2+4x$

h $-x(2x-4)$

$-2x^2+4x$

i $-x(4-2x)$

$-4x+2x^2$

j $x(x+4)$

x^2+4x

k $-3x(2-x)$

$-6x+3x^2$

l $3x(x-2)$

$3x^2-6x$

m $3x(x-2y)$

$3x^2-6xy$

n $3x(5x-2y)$

$15x^2-6xy$

o $-3x(2y-5x)$

$-6xy+15x^2$

p $-3x^2(2y-5x)$

$-6x^2y+15x^3$

q $-3y^2(2y-5x)$

$-6y^3+15xy^2$

r $-3y^2(2y-5xy)$

$-6y^3+15xy^3$

s $-3y^3(2-5x)$

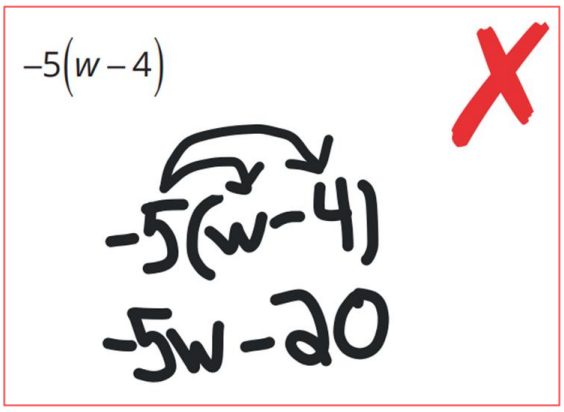
$-6y^3+15xy^3$

t $-3x(2-x)$

$-6x+3x^2$

3 Read the example carefully and answer the questions.

Destiny's work:



What did Destiny do wrong when applying the distributive property?

She didn't multiply …… and …… correctly.

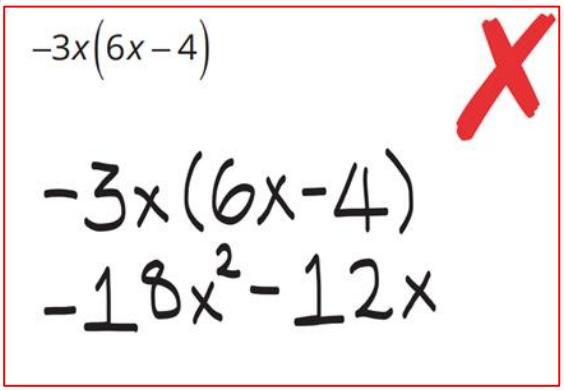
What should the correct answer be?

$-5w + 20$

Expand the following: $-2(w - 7)$

$-2w + 14$

Hugo's work



The correct answer is $-18x^2 + 12x$.
What rule did Hugo forget?

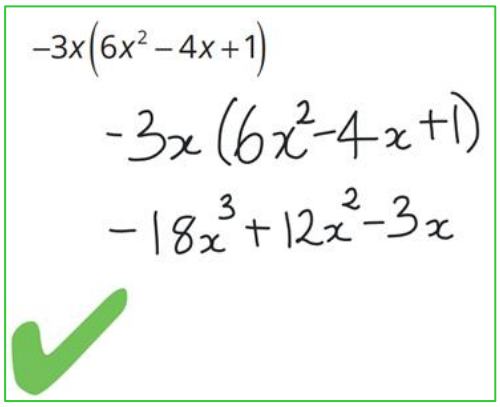
Negative times negative is ……

Expand:

$$-3x(6x^2 - 4x)$$

$-18x^3 + 12x^2$

Alicia's work



Why did Alicia write $-18x^3$ instead of $-18x^2$?

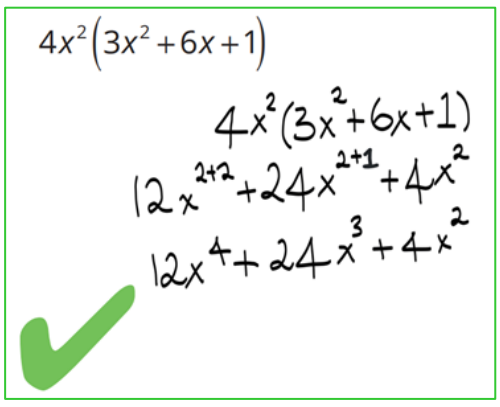
When you multiply x and x^2, you add the ……., so the answer is x^3

Expand:

$$-3x(6 - 4x + x^2)$$

$-18x + 12x^2 - 3x^3$

Mora's work



Why did Mora add the exponents instead of multiplying them?

You add the exponents when …… terms with the same base.

Expand:

$$-4x^2(3x^2 + 6x + 1)$$

$-12x^4 - 24x^3 - 4x^2$

4 Expand these expressions.

a	$2(x+3)$	**b**	$2(x-7)$	**c**	$-(2+x)$
	$2x-6$		$2x-14$		$-2-x$
d	$4(7-x)$	**e**	$6(2x+1)$	**f**	$9(2w+3h)$
	$28-4x$		$12x+6$		$18w+27h$
g	$-3(x+2)$	**h**	$-5(x-3)$	**i**	$-4(2-x)$
	$-3x-6$		$-5x+15$		$-8+4x$
j	$-20(9+x)$	**k**	$-5(2w-3)$	**l**	$-(w+9)$
	$-180-20x$		$-10w+15$		$-w-9$
m	$4(m-3n+5)$	**n**	$-2(p-3q-2)$	**o**	$2x(x+5)$
	$4m-12n+20$		$-2p+6q+4$		$2x^2+10x$
p	$6a(a-4)$	**q**	$-4x(3x-4y)$	**r**	$3y(5y+z-8)$
	$6a^2-24a$		$-12x^2+16xy$		$15y^2+3yz-24y$
s	$-2a(4b-7a+10)$	**t**	$7y(2y-2y^2-4)$	**u**	$-3a(2a^2-a-1)$
	$-8ab+14a^2-20a$		$14y^2-14y^3-28y$		$-6a^3+3a^2+4a$

5 Decide what goes in the box to make the following true.

a	$3(\square+2)=3x+6$	**b**	$3(\square+2)=3x+6$
	x		x
c	$5(k-\square)=5k-25$	**d**	$k(k-\square)=k^2-7k$
	5		7
e	$8(t-\square)=8t-32$	**f**	$-6(t-\square)=-6t+24$
	4		4
g	$\square(2x+5-y)=12x+30-6y$	**h**	$\square(2x+5-y)=2x^2+5x-xy$
	6		x
i	$\square(4-x-4y)=12-3x-12y$	**j**	$\square(4-x-4y)=-12+3x+12y$
	3		-3
k	$7(\square+10-\square)=7m+70-7n$	**l**	$7(\square+10-\square)=7m+70-14n$
	m,n		$m,2n$
m	$5(a-\square+\square)=5a-5b+25$	**n**	$5(a-\square+\square)=5a-20b+25$
	$b,5$		$4b,5$

Expanding then Simplifying

Review

1 Simplify these expressions.

a	$7a + 8 + 12a$	**b**	$12x - 8 + 9x - 13$	**c**	$-5xy + 55 - 7yx - 3$
d	$7a^2 + 3a - 12a$	**e**	$9y - 8x + 12xy + 9x$	**f**	$12xy^2 - 8y^2a + 13$

Expanding then Simplifying

After expanding brackets, **simplify** the result by combining **like terms**.

Example Expand then simplify the result

$4(3p + 2) - 5p$

$= 12p + 8 - 5p$

$= 7p + 8$

$5(2x - 3) - 4(1 - 5x)$

$= 10x - 15 - 4 + 20x$

$= 30x - 19$

Guided Practice

Expand, then simplify.

a $5(3m + 2) + 4m$

$= 15m + 10 + 4m$

$=$

$19m + 10$

b $3(1 - 5e) + 6e$

$3 - 9e$

c $4w - 2(5 + 2w)$

$= 4w - 10 - 4w$

$=$

-10

d $8 - 5(2x - 7)$

$43 - 10x$

e $5(2a + 3) + 4(a + 7)$

$14a + 43$

f $3(2d + 3) - 5(d + 4)$

$d - 11$

1 Expand the following and collect like terms.

a $3 + 2(x + 4)$

$11 + 2x$

b $2 + 5(3x - 1)$

$-3 + 15x$

c $3 + 4(x - 2)$

$-5 + 4x$

d $3 - 4(x - 2)$

$11 - 4x$

e $1 - 5(x + 4)$

$-19 - 5x$

f $5 - (x - 6)$

$11 - x$

g $9 - (x - 3)$

$12 - x$

h $5 - (3 + 2x)$

$2 - 2x$

i $7 + 2(x - 3)$

$1 + 2x$

2 Expand then simplify.

a $3(x - 4) + 1$

$3x - 11$

b $3(x - 4) - 2$

$3x - 14$

c $3(x - 4) - x$

$2x - 12$

d $3(x - 4) - 3x$

-12

e $2 + 3(x - 4)$

$3x - 10$

f $-3 + 3(x - 4)$

$3x - 15$

g $x + 3(x - 4)$

$4x - 12$

h $-2x + 3(x - 4)$

$x - 12$

i $3x(x - 4) + 2x$

$3x^2 - 10x$

j $2x + 3x(x - 4)$

$3x^2 - 10x$

k $-2x + 3x(x - 4)$

$3x^2 - 14x$

l $-3(x - 4) + 1$

$-3x + 13$

m $-3(x - 4) - 2$

$-3x + 10$

n $-3(x - 4) - x$

$-4x + 12$

o $-3(x - 4) - 3x$

$-6x + 12$

p $2 - 3(x - 4)$

$-3x + 14$

q $-3 - 3(x - 4)$

$-3x + 9$

r $x - 3(x - 4)$

$-2x + 12$

s $-2x - 3(x - 4)$

$-5x + 12$

t $-3x(x - 4) + 2x$

$-3x^2 + 14x$

u $2x - 3x(x - 4)$

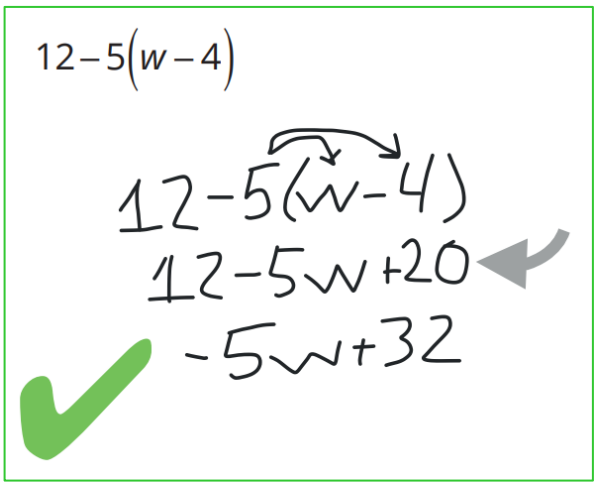
$-3x^2 + 14x$

v $-2x - 3x(x - 4)$

$-3x^2 + 10x$

3 Read the example carefully and answer the questions.

John's work:



Where did the +20 come from in the step marked with an arrow?

…… × …… = 20

Would he have gotten the same answer if he first subtracted 5 from 12? Explain.

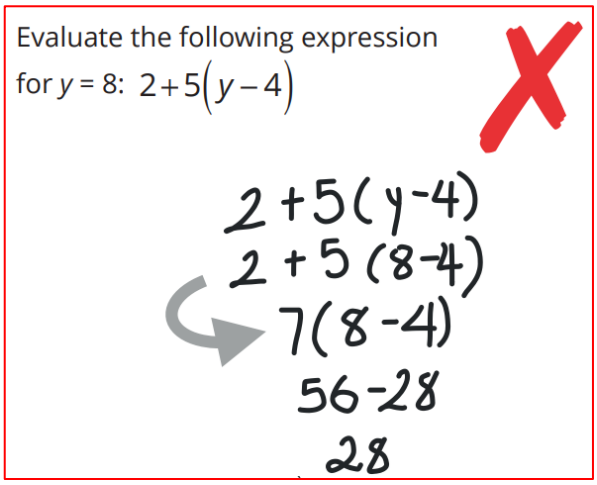
……., you would get 7w, which is incorrect as it doesn't follow the …… of ……

Expand and simplify

$$12 + 5(w - 4)$$

$5w - 8$

Louis' work



What did Louis do in the step marked with an arrow?

Added …… and ……

What should he have done instead?

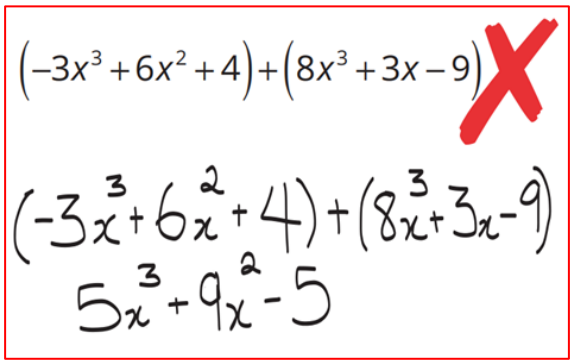
…… 4 from 8, or …… the brackets.

Evaluate the following expression for $y = 8$:

$$2y + 5(y - 4)$$

36

Eric's work



Eric made a mistake when adding the terms. What did he add incorrectly?

He added …… and ……., which are not …… terms.

Simplify:

$$(-3x^3 + 6x^2 + 4x) + (8x^3 - 3x + 9)$$

$5x^3 + 6x^2 + x + 9$

4 Expand and simplify.

a $3(x+2)+2(x+3)$

$5x+12$

b $3(x+2)+2(x-3)$

$5x$

c $3(x+2)-2(x+3)$

x

d $3(x+2)-2(x-3)$

$x+12$

e $p(p+2)+5(p+3)$

$p^2+7p+15$

f $p(p-3)+p(p+3)$

$2p^2$

g $p(3-2p)-3p(p-2)$

$-5p^2+9p$

5 Expand the following and collect like terms.

a $2(x+3)+3(x+2)$

$5x+12$

b $3(2x+1)+5(x-1)$

$11x-2$

c $4(3x+2)+5(x-3)$

$17x-7$

d $-2(x+2)+3(x-1)$

$x-7$

e $2(4x-3)-2(3x-1)$

$2x-4$

f $-(x+3)-3(x+5)$

$-4x-18$

g $-2(2x-4)-3(3x+5)$

$-13x-7$

h $3(3x-1)-2(2-x)$

$11x-7$

i $-4(5-x)-(2x-5)$

$2x-15$

6 Expand and simplify these expressions.

a $2x(x+1)+3x(x+4)$

$5x^2+14x$

b $3x(x+4)+2x(x+1)$

$5x^2+14x$

c $3x(x+1)+2x(x+4)$

$5x^2+11x$

d $3x(x-1)+2x(x+4)$

$5x^2+5x$

e $3x(x+1)+2x(x-4)$

$5x^2-5x$

f $3x(x-1)+2x(x-4)$

$5x^2-11x$

g $3x(x-1)-2x(x-4)$

x^2+5x

h $3x(x+1)-2x(x+4)$

x^2-5x

i $3x(x+1)-2(x+4)$

$3x^2+x-8$

j $3x(x+1)-5(x+4)$

$3x^2-2x-20$

k $3x^2(x+1)-5x(x+4)$

$3x^3-2x^2-20x$

l $3x^2(x-1)-5x(x-4)$

$3x^3-8x^2+20x$

m $3x^2(x-1)-(5x^2-4)$

$3x^3-8x^2+4$

n $3x^2(x-1)-5x^2$

$3x^3-8x^2$

o $5x^2-3x^2(x-1)$

$8x^2-3x^3$

p $5-3x^2(x-1)$

$5-3x^3+3x^2$

q $5+3x^2(1-x)$

$5+3x^2-3x^3$

r $5+3x^2(y-x)$

$5+3x^2y-3x^3$

7 Find the missing number to make these equations true.

a $4(k + \square) + 3(k - 5) = 7k - 3$

3

b $3(2 - r) + \square(6 - 2r) = r - 6$

-2

c $2(d - 5) - \square(1 - d) = 7d - 15$

5

8 Expand and simplify these expressions.

a $4x(3x + y) + 2x(5x - 2y)$

$22x^2$

b $5m(4m - 3n) - 2m(9m - 2n)$

$2m^2 - 11mn$

c $3a^2(4a^3 - b^4) + 2a^2(5a^3 + 3b^4)$

$22a^5 + 3a^2b^4$

d $2y^3(5y^2 + 3y^4) - 4y^2(2y^2 - y^4)$

$10y^5 + 6y^7 - 8y^4 + 4y^6$

e $5a^2b(2a^3b^2 - 3ab^3) - 4a^2b(3a^3b^2 - 4ab^3)$

$-2a^5b^3 + a^3b^4$

9 Find the missing expressions to make these equations true.

a $7(p - 3) - 4(\square) = 3p - 1$

$p - 5$

b $3(5p + 9) - 4(\square) = 3p - 1$

$3p + 7$

c $8(3p + 1) - 3(\square) = 3p - 1$

$7p + 3$

d $3(5p - 3) - 4(\square) = 3p - 1$

$3p - 2$

e $11(3p - 5) - 6(\square) = 3p - 1$

$5p - 9$

10 Identify the mistake in these expansions, then write out the correct expansion.

a $2(x + 6) = 2x + 6$

Mistake:

Didn't multiply 2 and 6

Correct answer:

$2x + 12$

b $x(x - 4) = 2x - 4x$

Mistake:

$x \times x = x^2$

Correct answer:

$x^2 - 4x$

c $-3(x + 4) = -3x + 12$

Mistake:

$-3 \times -4 = 12$

Correct answer:

$-3x - 12$

d $-7(x - 7) = -7x - 49$

Mistake:

$-7 \times -7 = +49$

Correct answer:

$-7x + 49$

e $5 - 2(x - 7) = 5 - 2x - 14$
$= -9 - 2x$

Mistake:

$-2 \times -7 = +14$

Correct answer:

$19 - 2x$

f $4(x - 2) - 3(x + 2) = 4x - 8 - 3x + 6$
$= x - 2$

Mistake:

$-3 \times 2 = -6$

Correct answer:

$x - 14$

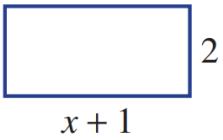
11 $5(2x + 1) + c(x + d) = 12x - 1$

What is the value of c and d?

$c = 2, d = -3$

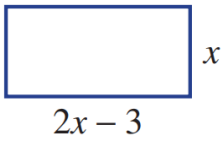
12 Find the area of these basic shapes in expanded form.

a



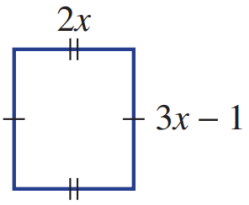
$2x + 2$

b



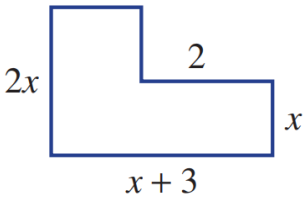
$2x^2 - 3x$

c



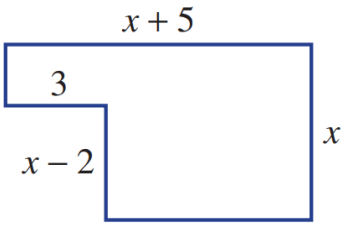
$6x^2 - 2x$

d



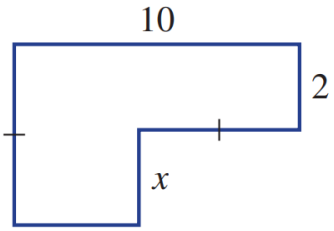
$2x^2 + 4x$

e



$6 + x^2 + 2x$

f



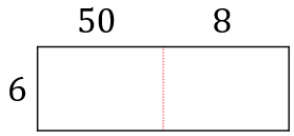
$20 + 8x - x^2$

Expanding Binomial Products

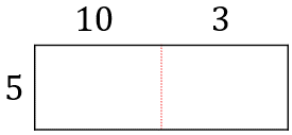
Area Model

Calculate the following using the area model. (do not use the algorithm!)

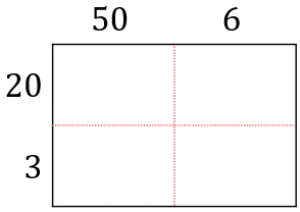
a $6 \times 58 = 6(50 + 8)$



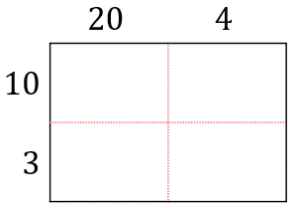
b $5 \times 13 = 5(10 + 3)$



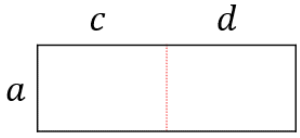
c $23 \times 56 = (20 + 3)(50 + 6)$



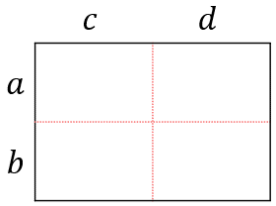
d $13 \times 24 = (10 + 3)(20 + 4)$



e $a(c + d)$



a $(a + b)(c + d)$



Binomial Products

A **binomial** has two terms in its simplest form: $a + b$

When you **multiply** two binomial expressions, this is called a **binomial product**.

We can expand binomial products using the area model.

$(a + b)(c + d)$

$= a(c + d) + b(c + d)$

$= ac + ad + bc + bd$

	c	d
a	ac	ad
b	bc	bd

Guided Practice

a $(p + 4)(q + 6)$

$= p(q + 6) + 4(q + 6)$

$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots$

	p	4
q		
6		

$pq + 4q + 6p + 24$

b $(x - 2)(x + 6)$

$= x(\quad) - 2(\quad)$

$=$

$=$

	x	6
x		
-2		

$x^2 + 4x - 12$

c $(2a - b)(3a + 2)$

	$3a$	2
$2a$		
$-b$		

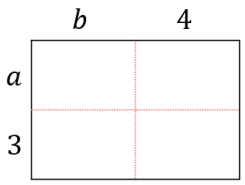
$6a^2 + 4a - 3ab - 2b$

d $(4x + 3)(2x + 5)$

$8x^2 + 26x + 15$

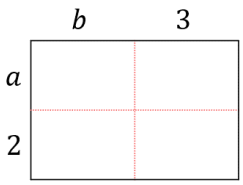
1 Expand these expressions. Use the area model to help you.

a $(a+3)(b+4)$



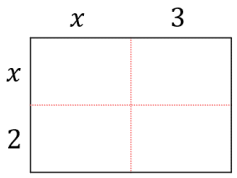
$ab+4a+3b+12$

b $(a+2)(b+3)$



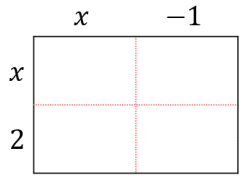
$ab+3a+2b+6$

c $(x+2)(x+3)$



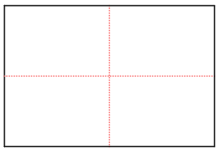
x^2+5x+6

d $(x+2)(x-1)$



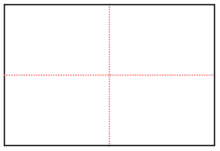
x^2+x-2

e $(2x+3)(x+1)$



$2x^2+5x+3$

f $(x-2)(2x+5)$



$2x^2+x-10$

2 Expand these expressions without an area model.

a $(x+1)(x+5)$

$= x(x+5)+1(x+5)$

$= \ldots\ldots\ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots\ldots\ldots$

x^2+6x+5

b $(x-3)(x+2)$

$= x(\quad)-3(\quad)$

$= \ldots\ldots\ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots\ldots\ldots$

x^2-x-6

c $(3x-2)(7x+2)$

$= 3x(\quad)-2(\quad)$

$= \ldots\ldots\ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots\ldots\ldots$

$21x^2-8x-4$

3 Expand these expressions. Set up an area model.

a $(x+y+5)(x+3)$

$x^2+xy+8x+3y+15$

b $(x^2+2x+3)(x-5)$

$x^3-3x^2-7x-15$

Expanding Binomial Products using the Rule

Expand Binomial Products using the Rule

1. Multiply each term in the first bracket by each term in the second bracket.
2. Simplify the result.

$$(a+b)(c+d)$$
$$= a(c+d) + b(c+d)$$
$$= ac + ad + bc + bd$$

Shortcut:

$$(a+b)(c+d) = ac + ad + bc + bd$$

Example Expand binomial products

$(x+3)(x+5)$

$$(x+3)(x+5) = x^2 + 5x + 3x + 15$$
$$= x^2 + 8x + 15$$

$(x-4)(x+7)$

$$(x-4)(x+7) = x^2 + 7x - 4x - 28$$
$$= x^2 + 3x - 28$$

Guided Practice

Expand and simplify these expressions.

a $(x+4)(x+2)$

$x^2 + 6x + 8$

b $(x-5)(x-2)$

$x^2 - 7x + 10$

c $(2x-1)(x-6)$

$2x^2 - 13x + 6$

d $(5x-2)(3x+7)$

$15x^2 + 29x - 14$

Key Ideas

1. We can multiply binomials using the rule:

$(a+b)(x+y) = a(\quad) + b(\quad)$

$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

1 Expand and simplify these expressions.

a $(x+5)(x+3)$

$x^2+8x+15$

b $(x+3)(x+5)$

$x^2+8x+15$

c $(x-3)(x+5)$

$x^2-2x-15$

d $(x+3)(x-5)$

$x^2-2x-15$

e $(x-3)(x-5)$

$x^2-8x+15$

f $(x-3)(x-3)$

x^2-6x+9

g $(x-3)^2$

x^2-6x+9

h $(x+3)^2$

x^2+6x+9

i $(x+6)^2$

$x^2+12x+36$

j $(x+6)(x-6)$

x^2-36

k $(x+5)(x-5)$

x^2-25

l $(x+5)(5-x)$

$-x^2+25$

m $(3x-1)(4x+4)$

$12x^2+8x-4$

n $(3x-1)(4x-4)$

$12x^2-16x+4$

o $(3x+2)(3x-2)$

$9x^2-4$

p $(3x-2)(3x-2)$

$9x^2-12x+4$

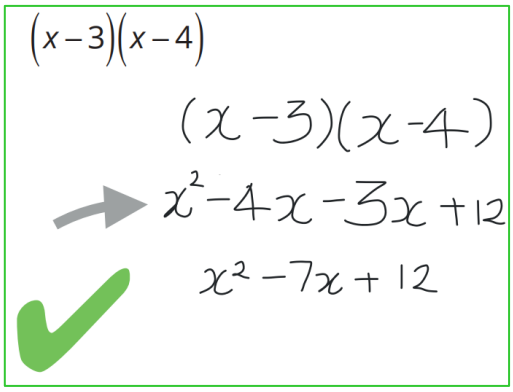
q $(ax-b)(cx-d)$

$acx^2-adx-bcx+bd$

r $(ax+b)(cx+d)$

$acx^2+adx+bcx+bd$

2 Read the example carefully and answer the questions.



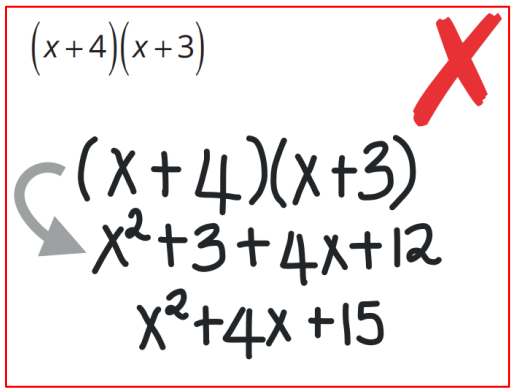
In the step marked with an arrow, where did the + sign come from?

..... × ..... = +12

Expand

$$(x-3)(x-6)$$

$x^2 - 9x + 18$



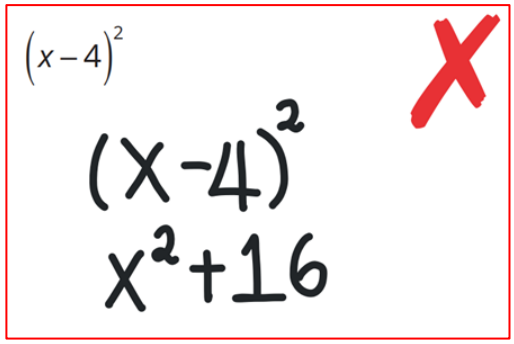
In the step marked with an arrow, what mistake did this student make?

The 3 should be ..... instead.

Expand

$$(x+4)(x-3)$$

$x^2 + x - 12$



What is $(x-4)^2$?

$(x-4)^2 = (\quad)(\quad)$

Expand

$$(x-4)^2$$

$x^2 + 8x + 16$

3

c Solve $2x + 8 = 10$

$x = 1$

d Factorise $2x + 8$

$2(x + 4)$

e Evaluate $2x + 8$ if $x = 3$

14

f Expand and simplify $(2x + 8)^2$

$4x^2 + 32x + 64$